■ 1.4.8 Picking Out Pieces of Algebraic Expressions

`Coefficient[`*expr*`,` *form*`]`	the coefficient of *form* in *expr*
`Exponent[`*expr*`,` *form*`]`	the maximum power of *form* in *expr*
`Numerator[`*expr*`]`	the numerator of *expr*
`Denominator[`*expr*`]`	the denominator of *expr*

Functions to pick out pieces of algebraic expressions.

Here is an algebraic expression.

```
In[1]:= e = Expand[(1 + 3x + 4y^2)^2]

                       2      2         2       4
Out[1]= 1 + 6 x + 9 x  + 8 y  + 24 x y  + 16 y
```

This gives the coefficient of `x` in `e`.

```
In[2]:= Coefficient[e, x]

                  2
Out[2]= 6 + 24 y
```

`Exponent[`*expr*`,` *y*`]` gives the highest power of *y* that appears in *expr*.

```
In[3]:= Exponent[e, y]
Out[3]= 4
```

`Coefficient` and `Exponent` are effectively functions for working with *polynomials*. They work only on expressions that are explicitly written out as sums of terms, in the form you get from `Expand`.

`Numerator` and `Denominator` work on rational expressions.

Here is a rational expression.

```
In[4]:= r = (1 + x)/(2 (2 - y))

          1 + x
Out[4]= ---------
        2 (2 - y)
```

`Denominator` picks out the denominator.

```
In[5]:= Denominator[%]
Out[5]= 2 (2 - y)
```

`Denominator` gives 1 for expressions that are not quotients.

```
In[6]:= Denominator[x^2]
Out[6]= 1
```

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